

44.3 Varying-coefficient models

The additive model of Definition 44.1.1 has no interactions: whatever z_2 does, the contribution of z_1 is the same curve. Throwing that restriction away brings back the curse of dimensionality, so relax it in one direction at a time. The most useful way keeps a *linear* effect and lets its slope move.

44.3.1 The model

Definition 44.3.1 — Varying-coefficient model

Let $u_1, \dots, u_n$ be the values of an **interaction variable** and $z_1, \dots, z_n$ those of an **effect modifier**. A **varying-coefficient term** is

$$f(z_i) u_i, \quad \sum_{i=1}^n f(z_i) = 0, \quad (44.3.1)$$

and the **varying-coefficient model** is a structured additive model (Definition 44.1.1) in which one or more of the terms have this form. Combined with a parametric column $\mathbf{u} = (u_1, \dots, u_n)^\top$ carrying a coefficient β_u , the total effect of $\mathbf{u}$ on the response is

$$E(y_i) = \dots + \{\beta_u + f(z_i)\} u_i, \quad (44.3.2)$$

so that $\beta_u + f(z)$ is the **coefficient function**: the slope in u at effect modifier value z . With $u_i \equiv 1$ the term is an ordinary smooth main effect, and with f constrained to be constant it is an ordinary linear effect.

Three readings of the same object:

- **An interaction**, the smooth analogue of the product term βuz of Section 5.7. Where a product term says the slope in u is *linear* in z , this one says only that it is smooth.
- **A factor-by-curve model**, when $\mathbf{u}$ indicates a group: each group gets its own deviation curve in z .
- **A time-varying coefficient**, when z is calendar time: the relationship has drifted, and the model does not say how.

44.3.2 Estimation is the same computation

Varying coefficients cost nothing new because the design of the term is a row-scaled basis.

Proposition 44.3.1 — A varying-coefficient term is an ordinary penalized term

Let $B_1, \dots, B_d$ be a basis for the effect modifier and $\mathbf{P}_0$ the penalty that the correspond-

ing smooth main effect would use. Put $(\mathbf{Z})_{il} = u_i B_l(z_i)$.

- (a) With $f = \sum_l \gamma_l B_l$, the vector of the term's contributions is $\{f(z_i)u_i\}_i = \mathbf{Z}\boldsymbol{\gamma}$, and the roughness of f is $\boldsymbol{\gamma}^\top \mathbf{P}_0 \boldsymbol{\gamma}$ as before. The term is therefore the pair $(\mathbf{Z}, \mathbf{P}_0)$, and everything in Sections 44.1 and 44.2 applies unchanged.
- (b) The constraint Eq. 44.3.1 is $\mathbf{c}^\top \boldsymbol{\gamma} = 0$ with $\mathbf{c} = (\sum_i B_1(z_i), \dots, \sum_i B_d(z_i))^\top$ — the *unweighted* column sums of the basis, not of $\mathbf{Z}$ — and is absorbed by Proposition 44.1.1(c). Under it, β_u in Eq. 44.3.2 is the average of the coefficient function over the design points, $\beta_u = n^{-1} \sum_i \{\beta_u + f(z_i)\}$.
- (c) If $u_i \neq 0$ for at least d design points with distinct z_i and the basis is linearly independent there, $\mathbf{Z}$ has full column rank. Observations with $u_i = 0$ contribute nothing to the term: the coefficient function is estimated only where the interaction variable is nonzero.

Proof.

- (a) The i th entry of $\mathbf{Z}\boldsymbol{\gamma}$ is $\sum_l \gamma_l u_i B_l(z_i) = u_i f(z_i)$. The penalty is a statement about f , so it is unchanged by multiplying the term by u_i .
- (b) $\sum_i f(z_i) = \sum_l \gamma_l \sum_i B_l(z_i) = \mathbf{c}^\top \boldsymbol{\gamma}$, which is the stated constraint; Proposition 44.1.1(c) applies verbatim with $\mathbf{c}$ in place of $\mathbf{Z}^\top \mathbf{1}$. The averaging statement is then $n^{-1} \sum_i \{\beta_u + f(z_i)\} = \beta_u + n^{-1} \mathbf{c}^\top \boldsymbol{\gamma} = \beta_u$.
- (c) Suppose $\mathbf{Z}\boldsymbol{\gamma} = \mathbf{0}$. Then $u_i f(z_i) = 0$ for all i , so $f(z_i) = 0$ at the d design points with $u_i \neq 0$; linear independence of the basis at those points forces $\boldsymbol{\gamma} = \mathbf{0}$. The last sentence is the same computation read forwards: rows with $u_i = 0$ are zero rows of $\mathbf{Z}$.

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Warning. Constraining the *weighted* sum $\sum_i u_i f(z_i) = 0$ instead — what centring the columns of $\mathbf{Z}$ does — also identifies the model, but β_u is then not the average slope, and if $\mathbf{u}$ has been centred it means nothing at all, since $\sum_i u_i = 0$ leaves the constraint silent about the level of f . Use (b).

44.3.3 An example, and how far it can be pushed

Example. (Does poverty matter more in urban states?)

Example (Poverty and murder rates) fitted the murder rate of the 50 US states in 2009 on the poverty rate, the single-parent percentage and the urban percentage, reporting one coefficient for poverty. Here that coefficient may vary smoothly with how urban the state is:

$$\text{murder}_i = \beta_0 + \beta_1 \text{single}_i + \{\beta_2 + f(\text{urban}_i)\} (\text{poverty}_i - \overline{\text{poverty}}) + f_0(\text{urban}_i) + \varepsilon_i,$$

with P-spline terms of 8 basis functions for f and f_0 and smoothing parameters chosen by the restricted-likelihood updates of Section 44.5. The fit uses 6.83 effective degrees of freedom in all, of which 1.82 go to the smooth main effect of urbanization and 2.01 to the coefficient function.

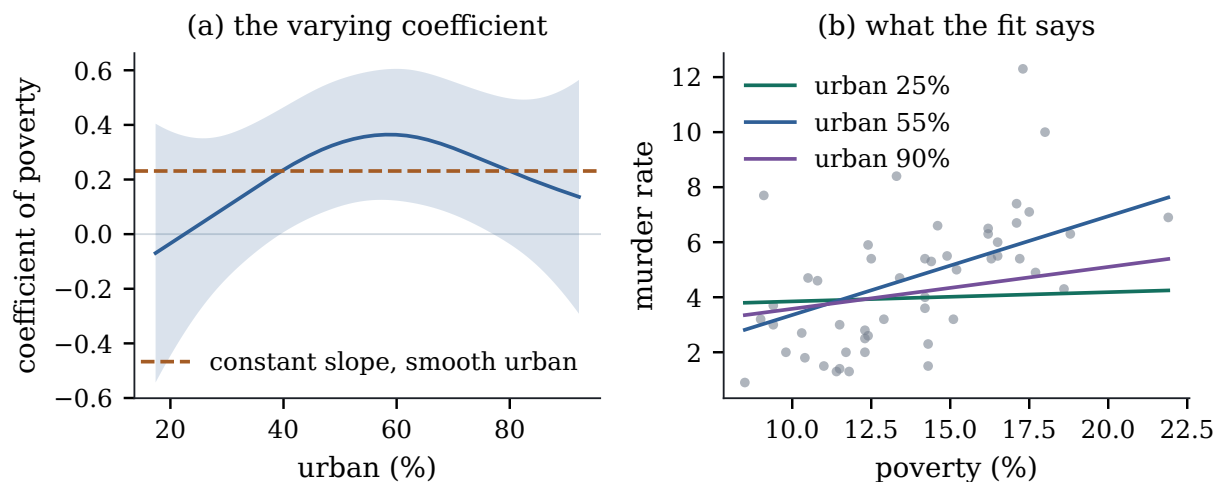


Figure 44.3.1: A varying coefficient on 50 US states. (a) the coefficient of poverty against the urban percentage, two standard errors wide; dashed, the constant slope from the model that keeps the smooth urban effect. (b) fitted murder rate against poverty at three levels of urbanization.

The average poverty slope is 0.250. Keeping the smooth main effect of urbanization but forcing the poverty slope to be constant gives 0.231, and Example (Poverty and murder rates), in which urbanization entered linearly as well, gave 0.2538. The estimated coefficient function rises from -0.034 at 20% urban to 0.364 at 60% and falls back to 0.152 at 90%: on its face, a story in which poverty is associated with homicide in middling states and not in the most rural ones.

That story does not survive its standard errors. The band of ± 2 standard errors in Figure 44.3.1 contains the constant coefficient over the whole range, and the contrast between 60% and 20% urban is 0.398 with a standard error of 0.234 , a ratio of 1.70 . Fifty states and a residual standard deviation of 1.418 murders per 100,000 cannot resolve a coefficient function; they can only say that one is not needed.

Fitting takes no new code, because Proposition 44.3.1(a) has made this an ordinary term: the design is the P-spline basis in the urban percentage scaled row by row by the centred poverty column, the penalty is the same second-difference matrix, and the same solver handles it. The band is read off the joint covariance of $(\beta_2, \boldsymbol{\gamma})$ with $\beta_2 + f(z)$ evaluated on a grid; the code is `code/ch44/varying_coefficient.py`.

Remark (What a coefficient function means). Example (Does poverty matter more in urban states?) is an observational comparison across states, and Chapter 25 applies to it word for word. A varying coefficient describes how an *association* changes with the modifier. Reading it as “raising poverty by one point in a 60%-urban state would raise the murder rate by 0.36” requires everything Theorem 25.6.2 requires, and requires it separately at each level of the modifier, since the confounding structure may differ across it. The extra flexibility buys a richer description, not weaker causal assumptions.

44.3.4 Where varying coefficients take you

Two extensions use nothing beyond Proposition 44.3.1. With z time and $\mathbf{u}$ a covariate observed repeatedly, $f(t_i)u_i$ is a coefficient drifting over the study; for series data the band must then

come from [Part VII](#) rather than from the independence assumption of Definition 44.1.1 ([Exercise \(B1\)](#)). With z a location and a spatial term from [Section 44.4](#) in place of the P-spline, the coefficient varies over a map, which is disease mapping: the same code with a different penalty.

When neither covariate deserves to be the linear one, a genuine bivariate term $f(z_1, z_2)$ is needed. The standard construction is a **tensor-product** basis $B_{lm}(z_1, z_2) = B_l^{(1)}(z_1)B_m^{(2)}(z_2)$ with the penalty

$$\mathbf{P} = \lambda_1(\mathbf{I} \otimes \mathbf{P}^{(1)}) + \lambda_2(\mathbf{P}^{(2)} \otimes \mathbf{I}), \quad (44.3.3)$$

penalizing roughness along each coordinate separately, and so invariant to the units of the two covariates. Everything in this chapter goes through for it, with two smoothing parameters instead of one. The book does not develop the construction further; how to scale the two penalties, how to separate the interaction from its main effects and how to display the surface belong to a book on smoothing, and Wood (2017) is where to find them. [Exercise \(C1\)](#) builds the smallest case by hand.

44.3.5 Exercises

44.3.5.1 A. Check your understanding

Exercise A1.

What does a varying-coefficient term become when (i) $u_i \equiv 1$; (ii) f is forced to be constant; (iii) f is forced to be linear in z ? Which of the three has a familiar name from [Part II](#)?

44.3.5.2 B. Practice

Exercise B1.

A varying-coefficient model is fitted to a time series, but the errors follow $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{V}$ with $\mathbf{V} \neq \mathbf{I}$. Show that the penalized estimate $\hat{\boldsymbol{\gamma}} = \mathbf{A}^{-1} \mathbf{Z}^\top \mathbf{y}$ is still unbiased for the penalized target $\mathbf{A}^{-1} \mathbf{Z}^\top \mathbf{E}(\mathbf{Y})$, but that its covariance is $\sigma^2 \mathbf{A}^{-1} \mathbf{Z}^\top \mathbf{V} \mathbf{Z} \mathbf{A}^{-1}$, not $\sigma^2 \mathbf{A}^{-1} \mathbf{Z}^\top \mathbf{Z} \mathbf{A}^{-1}$. If $\mathbf{V}$ has positive autocorrelations, in which direction is the naive band wrong?

Solution. Linearity gives both at once: $\mathbf{E}(\mathbf{A}^{-1} \mathbf{Z}^\top \mathbf{Y}) = \mathbf{A}^{-1} \mathbf{Z}^\top \mathbf{E}(\mathbf{Y})$ and $\text{Cov}(\mathbf{A}^{-1} \mathbf{Z}^\top \mathbf{Y}) = \mathbf{A}^{-1} \mathbf{Z}^\top \text{Cov}(\mathbf{Y}) \mathbf{Z} \mathbf{A}^{-1}$ (Theorem 2.2.1). Under positive autocorrelation $\mathbf{Z}^\top \mathbf{V} \mathbf{Z} - \mathbf{Z}^\top \mathbf{Z}$ is typically nonnegative definite for smooth columns, so the true variance exceeds the naive one and the band is too narrow, as Theorem 19.2.2 found for a linear model. The repair is [Chapter 31](#), or a random-effect term for the serial structure as in [Chapter 33](#).

Exercise B2.

Show that a model containing both a smooth main effect $f_0(z)$ and a varying-coefficient term $f(z)u$ has exact concurvity (Definition 44.2.1) as soon as u_i is itself a function of z_i , say $u_i = h(z_i)$ with h in the span of the basis. What does this say about the model

of [Example \(Does poverty matter more in urban states?\)](#)?

Solution. If $u_i = h(z_i)$ with h in the span, the varying term contributes $f(z_i)h(z_i)$, a function of z_i alone; whenever fh is again in the span, the main-effect term realizes it too, and subtracting gives a nontrivial relation. In [Example \(Does poverty matter more in urban states?\)](#) the poverty rate is *not* a function of the urban percentage, so the concurvity is partial rather than exact; but the two are correlated across states, so the terms compete, which is part of why the bands are wide.

44.3.5.3 C. Going deeper

Exercise C1.

Build the smallest tensor-product term by hand: take $d_1 = d_2 = 3$ basis functions in each coordinate and first-difference penalties $\mathbf{P}^{(1)} = \mathbf{P}^{(2)}$. Write out the 9×9 penalty of [Eq. 44.3.3](#) for $\lambda_1 = \lambda_2 = 1$ and find its null space. Why does that null space make the term identifiable only after the main effects have been constrained away?